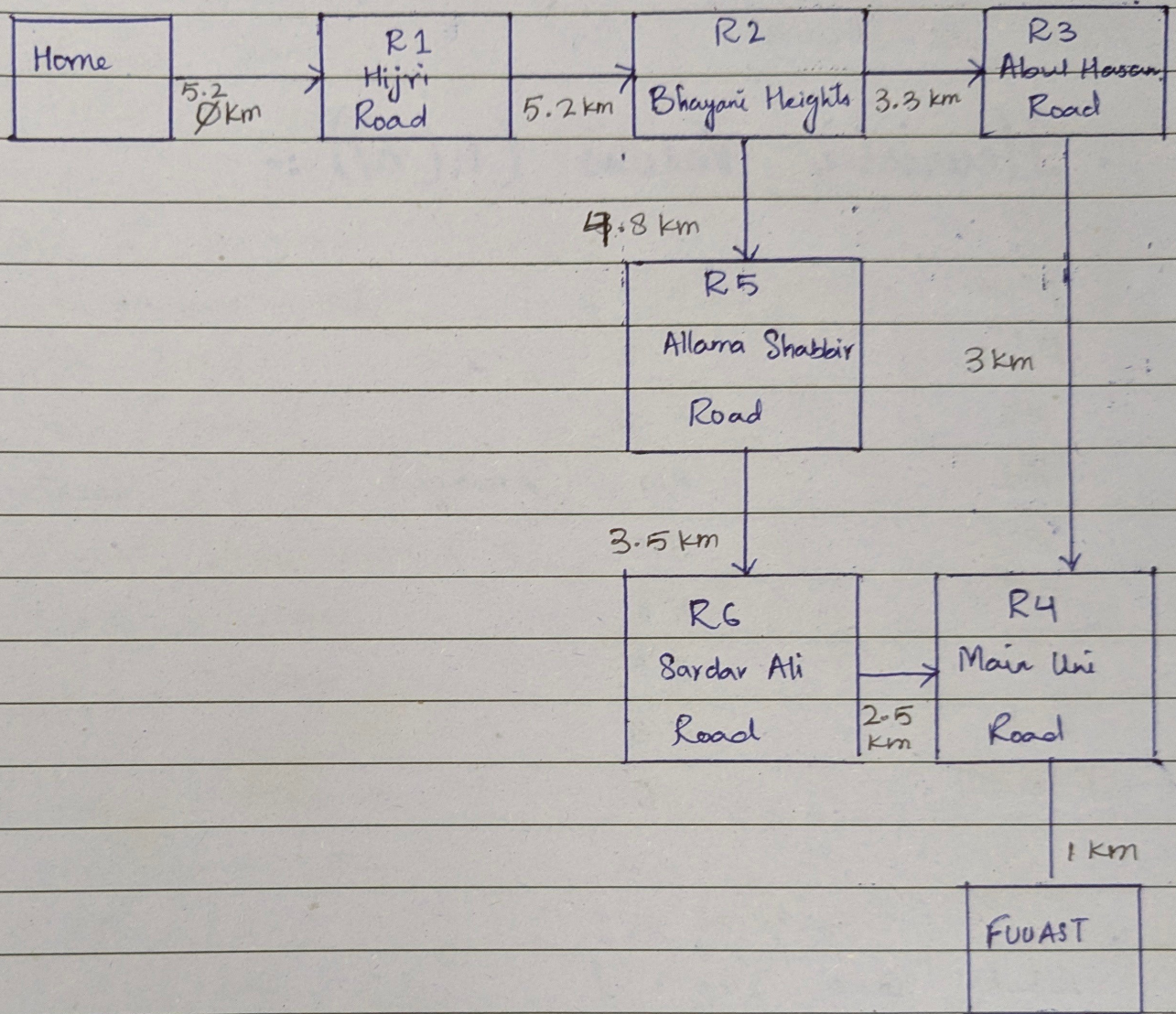


A* Search

Date _____

. Map :-



. Connections :-

- Home → R1
- R1 → R2
- R2 → R3, R5
- R3 → R4

- R5 $\rightarrow$ R6
- R6 $\rightarrow$ R4
- R4 $\rightarrow$ FUUAST

• Heuristic Values ($h(n)$) :-

Node	$h(n)$
Home	5.2
R1	5.2
R2	4.0 (Average)
R3	3.3
R5	4.8
R6	3.5
R4	1.0
FUUAST	0

Cost Assumption :-
Each road step = 1

• Search :-

Step 0 :- Initialize

PQ : [Home]

v : $\emptyset$

$$g = 0$$

$$f = 5.2$$

- Parent Table:-

Home $\rightarrow$ None

Step 1: Expand Home

Neighbours : R1

PQ : [R1]

v : [Home]

Node	g	h	f
R1	1	5.2	6.2

- Choose R1 ($f = 6.2$)
- Parent Table Update :-

Home $\rightarrow$ None

R1 $\rightarrow$ Home

Step 2: Expand R1

Neighbours : R2

PQ : [R2]

v : [Home, R1]

Node	g	h	f
R2	2	4.0	6.0

- Choose R2 ($f = 6.0$)
- Parent Table Update :-

Home $\rightarrow$ None

R1 $\rightarrow$ Home

R2 $\rightarrow$ R1

Step 3 : Expand R2

Neighbours: R3, R5

PQ: [R3, R5]

V: [Home, R1, R2]

Node	g	h	f
R3	3	3.3	6.3
R5	3	4.8	7.8

- Choose R3 ($f=6.3$)
- Parent Table Update:-

Home $\rightarrow$ None

R1 $\rightarrow$ Home

R2 $\rightarrow$ R1

R3 $\rightarrow$ R2

R5 $\rightarrow$ R2

Step 4 : Expand R3

Neighbours: R4

PQ: [R4, R5]

V: [Home, R1, R2, R3]

Node	g	h	f
R4	4	1.0	5.0
R5	3	4.8	7.8

- Choose R4 ($f=5.0$) (from S:3)

- Parent Table Update:-

Home $\rightarrow$ None

R1 $\rightarrow$ Home

R2 $\rightarrow$ R1

R3 $\rightarrow$ R2

R5 $\rightarrow$ R2

R4 $\rightarrow$ R3

Step 5: Expand R4

Neighbours: FUUAST

PQ: [FUUAST, R5]

γ : [Home, R1, R2, R3, R4]

Node	g	h	f
FUUAST	5	0	5.0
R5	3	4.8	7.8

• Choose FUUAST ($f=5.0$)

• Parent Table Update:-

Home $\rightarrow$ None

R1 $\rightarrow$ Home, R2 $\rightarrow$ R1

R3 $\rightarrow$ R2

R5 $\rightarrow$ R2

R4 $\rightarrow$ R3

FUUAST $\rightarrow$ R4

Step 6: Goal Found

• Final Parent Table

Node	Parent
Home	None
R1	Home
R2	R1
R3	R2
R5	R2
R4	R3
FUUAST	R4

- Start from goal and go backward:

FUVAST $\rightarrow$ R4 $\rightarrow$ R3 $\rightarrow$ R2 $\rightarrow$ R1 $\rightarrow$ Home

- Reverse It:

Home $\rightarrow$ R1 $\rightarrow$ R2 $\rightarrow$ R3 $\rightarrow$ R4 $\rightarrow$ FUVAST